

PSEUDOCODE WRITING

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QUESTIONS

Q1

 Student Management System [BL3 - Apply]
10 marks

Q2

 Library Book Management [BL3 - Apply]
10 marks

Q3

 Employee Salary Calculation [BL3 - Apply]
10 marks

Q4

 Shape Area Calculator using Polymorphism [BL3 - Apply]
10 marks

Q5

 5. Vehicle Registration System [BL3 - Apply]
10 marks

Q6

 6. Bank Account Hierarchy [BL4 - Analyse]
10 marks

Q7

 7. Hospital Patient Record System [BL4 - Analyse]
10 marks

Q8

 8. E-Commerce Product Catalog [BL4 - Analyse]
10 marks

Q9

 9. University Staff Hierarchy [BL4 - Analyse]
10 marks

Q10

 10. Smart Home Device Controller [BL4 - Analyse]
10 marks

10/10 answered

Q1 Student Management System [BL3 - Apply]

10 marks

Write pseudocode to design a Student class with attributes (name, rollNo, marks) and methods to:

- Accept student details
- Calculate average marks
- Display result (Pass/Fail)

Apply encapsulation and data hiding by restricting direct access to attributes.

OOP Concepts: Encapsulation | Data Hiding | Classes & Objects

Your Answer

```

CLASS Student
    PRIVATE name
    PRIVATE rollNo
    PRIVATE marks[3]
    PRIVATE average

    METHOD acceptDetails()
        INPUT name
        INPUT rollNo
        FOR i = 1 TO 3
            INPUT marks[i]
        END FOR
    END METHOD

    METHOD calculateAverage()
        sum = 0
        FOR i = 1 TO 3
            sum = sum + marks[i]
        END FOR
        average = sum / 3
    END METHOD

    METHOD displayResult()
        IF average >= 40 THEN
            PRINT "Pass"
        ELSE
            PRINT "Fail"
        END IF
    END METHOD
END CLASS

START
    CREATE Student s
    s.acceptDetails()
    s.calculateAverage()
    s.displayResult()
END
  
```

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Q7 ✓
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System [BL4 - Analyze]
10 marks

Q8 ✓
8. E-Commerce Product
Catalog [BL4 - Analyze]
10 marks

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[BL4 - Analyze]
10 marks

Q10 ✓
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Controller [BL4 - Analyze]
10 marks

10/10 answered

Q2 Library Book Management [BL3 - Apply]

10 marks

Design a Book class with private attributes (title, author, ISBN, isAvailable). Implement methods to:

- Issue a book (change availability status)
- Return a book
- Display book details using getter methods

Demonstrate object creation for at least 3 books and simulate issue/return operations.

OOP Concepts: Encapsulation | Getters & Setters | Objects

Your Answer

```

PRIVATE title
PRIVATE ISBN
PRIVATE isAvailable

METHOD setDetails(t, a, i)
    title = t
    author = a
    ISBN = i
    isAvailable = TRUE
END METHOD

METHOD issueBook()
    IF isAvailable = TRUE THEN
        isAvailable = FALSE
        PRINT "Book Issued"
    ELSE
        PRINT "Book Not Available"
    END IF
END METHOD

METHOD returnBook()
    isAvailable = TRUE
    PRINT "Book Returned"
END METHOD

METHOD displayDetails()
    PRINT title, author, ISBN, isAvailable
END METHOD

END CLASS

START
    CREATE Book b1, b2, b3

    b1.setDetails("Java", "James", "101")
    b2.setDetails("Python", "Guido", "102")
    b3.setDetails("C++", "Bjame", "103")

    b1.issueBook()
    b1.returnBook()

    b1.displayDetails()
    b2.displayDetails()
    b3.displayDetails()

END
    
```

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10 marks

10/10 answered

Q3 Employee Salary Calculation [BL3 - Apply]

10 marks

Write pseudocode for an Employee class with encapsulated attributes (empId, name, basicPay, designation). Implement methods to:

- Calculate gross salary (basic + HRA + DA)
- Calculate net salary after deductions (PF, tax)
- Display pay slip

Apply data hiding so salary components are not directly accessible.
OOP Concepts: Encapsulation | Data Hiding | Methods

Your Answer

```

CLASS Employee
    PRIVATE empId, name, basicPay, designation

    METHOD calculateSalary()
        HRA = basicPay * 0.20
        DA = basicPay * 0.10
        gross = basicPay + HRA + DA
        net = gross - (basicPay * 0.12) - (gross * 0.05)
    END METHOD

    METHOD display()
        PRINT empId, name, designation, gross, net
    END METHOD
END CLASS

START
    CREATE Employee e
    e.calculateSalary()
    e.display()
END
    
```

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10/10 answered

Q4 Shape Area Calculator using Polymorphism [BL3 - Apply]

10 marks

Create a Shape base class with an overridable calculateArea() method. Apply this to:

- Circle — area using radius
- Rectangle — area using length and breadth
- Triangle — area using base and height

Use method overriding to demonstrate runtime polymorphism. Call calculateArea() for each object.

OOP Concepts: Inheritance | Polymorphism | Method Overriding

Your Answer

```

CLASS Shape
    METHOD calculateArea()
END METHOD
END CLASS

CLASS Circle EXTENDS Shape
    METHOD calculateArea()
        PRINT 3.14 * radius * radius
    END METHOD
END CLASS

CLASS Rectangle EXTENDS Shape
    METHOD calculateArea()
        PRINT length * breadth
    END METHOD
END CLASS

CLASS Triangle EXTENDS Shape
    METHOD calculateArea()
        PRINT 0.5 * base * height
    END METHOD
END CLASS

START
    CREATE Circle c
    CREATE Rectangle r
    CREATE Triangle t

    c.calculateArea()
    r.calculateArea()
    t.calculateArea()
END
    
```

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10/10 answered

Q5 5. Vehicle Registration System [BL3 - Apply]

10 marks

Design a Vehicle class with attributes (regNo, owner, fuelType). Extend it into:

- TwoWheeler — with engine capacity
- FourWheeler — with seating capacity and AC status

Apply inheritance to reuse common attributes and override a displayDetails() method in each subclass.

OOP Concepts: Inheritance | Method Overriding | Subclass

Your Answer

```

CLASS Vehicle
    regNo, owner, fuelType

    METHOD displayDetails()
        PRINT regNo, owner, fuelType
    END METHOD
END CLASS

CLASS TwoWheeler EXTENDS Vehicle
    engineCapacity

    METHOD displayDetails()
        PRINT regNo, owner, fuelType, engineCapacity
    END METHOD
END CLASS

CLASS FourWheeler EXTENDS Vehicle
    seatingCapacity, ACstatus

    METHOD displayDetails()
        PRINT regNo, owner, fuelType, seatingCapacity, ACstatus
    END METHOD
END CLASS

START
    CREATE TwoWheeler t
    CREATE FourWheeler f
    t.displayDetails()
    f.displayDetails()
END
    
```

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Q10
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10 marks

10/10 answered

Q6 6. Bank Account Hierarchy [BL4 - Analyze]

10 marks

Develop pseudocode for a BankAccount superclass and two subclasses — SavingsAccount (with interest calculation) and CurrentAccount (with overdraft facility). Analyze and implement:

- Inheritance of common attributes (accNo, balance, holder)
- Method overriding for withdrawal behavior
- How overdraft protection differs from savings limit

Compare the behavior of withdraw() across both subclasses and justify the design choices.
OOP Concepts: Inheritance | Method Overriding | Polymorphism

Your Answer

```

CLASS BankAccount
    accNo, balance, holder

    METHOD withdraw(amount)
        balance = balance - amount
    END METHOD
END CLASS

CLASS SavingsAccount EXTENDS BankAccount
    METHOD interest()
        balance = balance + balance * 0.04
    END METHOD

    METHOD withdraw(amount)
        IF balance - amount >= 500 THEN
            balance = balance - amount
        END IF
    END METHOD
END CLASS

CLASS CurrentAccount EXTENDS BankAccount
    METHOD withdraw(amount)
        IF balance - amount >= -5000 THEN
            balance = balance - amount
        END IF
    END METHOD
END CLASS

START
    CREATE SavingsAccount s
    CREATE CurrentAccount c
    s.withdraw(1000)
    c.withdraw(1000)
END
    
```

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Q10 ✓
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10 marks

10/10 answered

Q7 7. Hospital Patient Record System [BL4 - Analyze]

10 marks

Design a Patient base class and analyze how it extends into:

- InPatient — with ward number, admission date, and daily charges
- OutPatient — with appointment date and consultation fee

Analyze encapsulation requirements for sensitive data (diagnosis, prescription). Override a generateBill() method for each type and compare the billing logic differences. OOP Concepts: Encapsulation | Inheritance | Polymorphism

Your Answer

```
CLASS Patient
    PRIVATE name, diagnosis, prescription

    METHOD generateBill()
    END METHOD
END CLASS
```

```
CLASS InPatient EXTENDS Patient
    wardNo, admissionDate, dailyCharges

    METHOD generateBill()
        PRINT dailyCharges * days
    END METHOD
END CLASS
```

```
CLASS OutPatient EXTENDS Patient
    appointmentDate, consultationFee

    METHOD generateBill()
        PRINT consultationFee
    END METHOD
END CLASS
```

```
START
    CREATE InPatient i
    CREATE OutPatient o
    i.generateBill()
    o.generateBill()
END
```

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10 marks

Q9
9. University Staff Hierarchy
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10 marks

Q10
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10 marks

10/10 answered

Q8 8. E-Commerce Product Catalog [BL4 - Analyze]

10 marks

Analyze and design a Product class hierarchy for an e-commerce system:

- Electronics — with warranty, brand, voltage
- Clothing — with size, fabric, gender
- Grocery — with expiryDate and weight

Override an applyDiscount() method for each category with different discount rules. Analyze why polymorphism makes the cart checkout process flexible and extensible. OOP Concepts: Polymorphism | Inheritance | Method Overriding

Your Answer

```

CLASS Product
    price

    METHOD applyDiscount()
    END METHOD
END CLASS

CLASS Electronics EXTENDS Product
    warranty, brand, voltage

    METHOD applyDiscount()
        RETURN price * 0.90
    END METHOD
END CLASS

CLASS Clothing EXTENDS Product
    size, fabric, gender

    METHOD applyDiscount()
        RETURN price * 0.80
    END METHOD
END CLASS

CLASS Grocery EXTENDS Product
    expiryDate, weight

    METHOD applyDiscount()
        RETURN price * 0.95
    END METHOD
END CLASS

START
    CREATE Electronics e
    CREATE Clothing c
    CREATE Grocery g

    PRINT e.applyDiscount()
    PRINT c.applyDiscount()
    PRINT g.applyDiscount()
END
    
```


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10 marks

Q5
5. Vehicle Registration System [BL2 - Apply]
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Q6
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10 marks

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10 marks

Q8
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10 marks

Q9
9. University Staff Hierarchy [BL4 - Analyze]
10 marks

Q10
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10 marks

10/10 answered

Q9 9. University Staff Hierarchy [BL4 - Analyze]

10 marks

Analyze the design of a Staff base class extended by TeachingStaff and NonTeachingStaff. Further extend TeachingStaff into Professor and Lecturer (multilevel inheritance). For each level:

- Identify which attributes should be private vs protected
- Override calculateAllowance() at each level
- Analyze how protected access enables inheritance without full exposure. Justify the use of multilevel inheritance and explain how access modifiers enforce data hiding.

OOP Concepts: Multilevel Inheritance | Access Modifiers | Data Hiding

Your Answer

```

CLASS Staff
    PRIVATE id, name
    PROTECTED salary
    METHOD calculateAllowance()
    END METHOD
END CLASS

CLASS TeachingStaff EXTENDS Staff
    METHOD calculateAllowance()
        RETURN salary * 0.20
    END METHOD
END CLASS

CLASS NonTeachingStaff EXTENDS Staff
    METHOD calculateAllowance()
        RETURN salary * 0.10
    END METHOD
END CLASS

CLASS Professor EXTENDS TeachingStaff
    METHOD calculateAllowance()
        RETURN salary * 0.30
    END METHOD
END CLASS

CLASS Lecturer EXTENDS TeachingStaff
    METHOD calculateAllowance()
        RETURN salary * 0.25
    END METHOD
END CLASS
    
```

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10 marks

Q10
10. Smart Home Device
Controller [BL4 - Analyze]
10 marks

10/10 answered

Q10 10. Smart Home Device Controller [BL4 - Analyze]

10 marks

Analyze and design a SmartDevice superclass with subclasses: SmartLight, SmartThermostat, and SmartLock. Override an executeCommand(cmd) method in each, where the same command (e.g., 'on', 'off', 'status') produces device-specific behavior. Analyze:

- How polymorphism allows a single controller loop to manage all devices
- Security implications of data hiding in SmartLock (PIN must never be exposed) Design the class hierarchy and explain the OOP principles applied at each level.

OOP Concepts: Polymorphism | Encapsulation | Inheritance

Your Answer

```
CLASS SmartDevice
    PRIVATE PIN

    METHOD executeCommand(cmd)
    END METHOD
END CLASS

CLASS SmartLight EXTENDS SmartDevice
    METHOD executeCommand(cmd)
        PRINT "Light", cmd
    END METHOD
END CLASS

CLASS SmartThermostat EXTENDS SmartDevice
    METHOD executeCommand(cmd)
        PRINT "Thermostat", cmd
    END METHOD
END CLASS

CLASS SmartLock EXTENDS SmartDevice
    METHOD executeCommand(cmd)
        PRINT "Lock", cmd
    END METHOD
END CLASS

START
    CREATE SmartLight l
    CREATE SmartThermostat t
    CREATE SmartLock k

    l.executeCommand("on")
    t.executeCommand("on")
    k.executeCommand("on")
END
```

Save Answer